

E.G.S. PILLAY ENGINEERING COLLEGE
(Autonomous)

NAGAPATTINAM – 611002

(Affiliated to Anna University, Chennai | Accredited by NAAC with 'A++' Grade | Accredited by NBA TI(B.E. – CSE, CIVIL, ECE, EEE, MECH& B.Tech – IT,) | Approved by AICTE, New Delhi)



MASTER OF COMPUTER APPLICATIONS R - 2024
FIRST YEAR
CURRICULUM AND SYLLABUS FOR FIRST SEMESTER

Course Code	Course Name	L	T	P	C	Maximum Marks			Category
						CIA	ES	Total	
Theory Course									
2401CA101	Applied Probability and Statistics	3	1	-	4	40	60	100	FC
2402CA102	Advanced Data structures and Algorithms	3	-	-	3	40	60	100	PC
2402CA103	Computer Communication and Networks	3	-	-	3	40	60	100	PC
2402CA104	Human Resource Management	3	-	-	3	40	60	100	PC
2402CA105	Python Programming	3	-	-	3	40	60	100	PC
2402CA106	Object Oriented Software Engineering	3	-	-	3	40	60	100	PC
	Audit CourseI *	2	-	-	-	100	-	100	AC
Laboratory Course									
2402CA107	Advanced Data structures and Algorithms Laboratory	-	-	4	2	60	40	100	PC
2402CA108	Python Programming Laboratory	-	-	4	2	60	40	100	PC
2402CA109	Professional Development Course-I	-	-	2	1	100	0	100	EEC
Total		20	1	10	24	560	440	1000	

L-Lecture | T –Tutorial | P- Practical | CA – Continuous Assessment | ES – End Semester

2402CA101

APPLIED PROBABILITY AND STATISTICS

L	T	P	C
3	2	0	4

PREREQUISITE:

Basic Multivariable Calculus & Elementary Linear Algebra

COURSE OBJECTIVES:

1. To encourage students to develop a working knowledge of the central ideas of Linear Algebra
2. To enable students to understand the concepts of Probability and Random Variables.
3. To understand the basic probability concepts with respect to two dimensional random variables along with the relationship between the random variables and the significance of the central limit theorem.
4. To apply the small / large sample tests through Tests of hypothesis
5. To enable the students to use the concepts of multivariate normal distribution and principal components analysis

COURSE OUTCOMES:

CO1: Apply the concepts of Linear Algebra to solve practical problems.

CO2: Use the ideas of probability and random variables in solving engineering problems.

CO3 : Be familiar with some of the commonly encountered two dimension random variables and be equipped for a possible extension to multivariate analysis.

CO4: Use statistical tests in testing hypothesis on data.

CO5: Develop critical thinking based on empirical evidence and the scientific approach to knowledge development

COs Vs POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	-	-	-	-	-	-	1	3
CO2	3	3	-	-	-	-	-	1	1	3
CO3	3	3	-	-	-	-	-	1	1	3
CO4	3	3	-	-	-	-	-	-	1	3
CO5	2	2	-	-	-	-	-	2	1	3

UNIT I LINEAR ALGEBRA

12 Hours

Vector spaces – norms – Inner Products – Eigenvalues using QR transformations – QR factorization – generalized eigenvectors – Canonical forms – singular value decomposition and applications – pseudo inverse – least square approximations

UNIT II PROBABILITY AND RANDOM VARIABLES

12 Hours

Probability – Axioms of probability – Conditional probability – Bayes theorem – Random variables – Probability function – Moments – Moment generating functions and their properties – Binomial, Poisson, Geometric, Uniform, Exponential, Gamma and Normal distributions – Function of a random variable.

UNIT III TWO DIMENSIONAL RANDOM VARIABLES

12 Hours

Joint distributions – Marginal and conditional distributions – Functions of two dimensional random variables – Regression curve – Correlation.

UNIT IV TESTING OF HYPOTHESIS

12 Hours

Sampling distributions – Type I and Type II errors – Small and Large samples – Tests based on Normal, t, Chi square and F distributions for testing of mean, variance and proportions – Tests for independence of attributes and goodness of fit.

UNIT V MULTIVARIATE ANALYSIS

12 Hours

Random vectors and matrices – Mean vectors and covariance matrices – Multivariate normal density and its properties – Principal components – Population principal components – Principal components from standardized variables

TOTAL : 60 PERIODS

REFERENCES:

1. Dallas E Johnson, "Applied multivariate methods for data Analysis", Thomson and Duxbury press, Singapore, 1998.
2. Richard A. Johnson and Dean W. Wichern, "Applied multivariate statistical Analysis", Pearson Education, Fifth Edition, 6th Edition, New Delhi, 2013.
3. Bronson, R., "Matrix Operation" Schaum's outline series, Tata McGrawHill, New York, 2011
4. Oliver C. Ibe, "Fundamentals of Applied probability and Random Processes", Academic Press, Boston, 2014.
5. Johnson R. A. and Gupta C.B., "Miller and Freund's Probability and Statistics for Engineers", Pearson India Education, Asia, 9th Edition, New Delhi, 2017.

2402CA102

ADVANCED DATA STRUCTURES AND ALGORITHMS

L	T	P	C
3	0	0	3

Prerequisite Problem Solving And Programming

COURSE OBJECTIVES

1. To understand the linear and non linear data structures available in solving problems
2. To learn and use hierarchical data structures and its operations
3. To learn the usage of graphs and its applications
4. To select and design data structures and algorithms that is appropriate for problems
5. To study about NP Completeness of problems

COURSE OUTCOMES

At the end of the course, students will be able to

- CO1 Apply suitable data structures and algorithms to solve computational problems.
- CO2 Implement efficient data structures and algorithms for solving real-world computing tasks.
- CO3 Apply graph algorithms and string-matching techniques to real-life problem scenarios.
- CO4 Design and analyze algorithms using dynamic programming, greedy, and divide-and-conquer strategies.
- CO5 Evaluate and apply suitable design strategies for solving computational problems effectively

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	1	2	1	–	–	–	2	2
CO2	3	3	2	3	2	–	–	–	3	3
CO3	3	3	3	2	1	–	–	–	3	3
CO4	3	3	3	2	–	–	–	1	3	2
CO5	3	3	3	2	2	–	–	1	3	3
Avg	3	2.8	2.4	2.2	1.5	–	–	1	2.8	2.6

UNIT I LINEAR DATA STRUCTURES

9 Hours

Introduction – Arrays – Structures- Abstract Data Types (ADT)- Stack- Representing Stacks- Applications of stack – Infix to postfix conversion – evaluation of expression- Queue- Representing Queue- Applications of Queue- Linked Lists – singly Linked list- Doubly Linked lists

UNIT II HIERARCHICAL DATA STRUCTURES , SORTING AND INDEXING

9 Hours

Binary Search Trees: Basics – Insertion and Deletion-Tree Traversal- Red Black trees: Properties of Red-Black Trees – Rotations – Insertion – Deletion -B-Trees: Definition of B - trees – Basic operations on B-Trees – Deleting a key from a B-Tree- Sorting – Bubble sort - Quick Sort - Insertion Sort – Heap sort – Hashing - Hashing functions - Collision Resolution Techniques - Separate chaining - Open addressing

UNIT III GRAPHS

9 Hours

Elementary Graph Algorithms: Representations of Graphs – Breadth-First Search – Depth-First Search – Topological Sort – Strongly Connected Components- Minimum Spanning Trees: Growing a Minimum Spanning Tree – Kruskal and Prim- Single- Source Shortest Paths: The Bellman-Ford algorithm – Single-Source Shortest paths in Directed Acyclic Graphs – Dijkstra's Algorithm - Topological sort

UNIT IV ALGORITHM DESIGN TECHNIQUES,

9 Hours

Notations -Dynamic Programming: Matrix-Chain Multiplication – Elements of Dynamic Programming – Longest Common Subsequence- Greedy Algorithms: – Elements of the Greedy Strategy- An Activity-Selection Problem - Huffman Coding- Algorithms – Algorithms as a Technology -Time and Space complexity of algorithms- Asymptotic Analysis-Average and worst- case analysis-Asymptotic notation-Importance of efficient algorithms- Program performance measurement -Recurrences:– types The Recursion

UNIT V NP COMPLETE AND NP HARD

9 Hours

NP-Completeness: Polynomial Time – Polynomial-Time Verification – NP- Completeness and Reducibility– NP-Completeness Proofs – NP-Complete Problems.

TOTAL : 45 PERIODS

REFERENCES:

1. S.Sridhar,” Design and Analysis of Algorithms”, Oxford University Press, 1st Edition, 2014.
2. Adam Drozdex, “Data Structures and Algorithms in C++”, Cengage Learning, 4th Edition, 2013.
3. T.H. Cormen, C.E.Leiserson, R.L. Rivest and C.Stein, "Introduction to Algorithms", Prentice Hall of India, 3rd Edition, 2012.
4. Mark Allen Weiss, “Data Structures and Algorithms in C++”, Pearson Education, 3rd Edition, 2009.
5. E. Horowitz, S. Sahni and S. Rajasekaran, “Fundamentals of Computer Algorithms”, University Press, 2nd Edition, 2008.
6. Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman, “Data Structures and Algorithms”, Pearson Education, Reprint 2006.
7. T.H. Cormen, C.E.Leiserson, R.L. Rivest and C.Stein, "Introduction to Algorithms", Prentice Hall of India, 3rd Edition, 2012.
8. Mark Allen Weiss, “Data Structures and Algorithms in C++”, Pearson Education, 3rd Edition, 2009.
9. E. Horowitz, S. Sahni and S. Rajasekaran, “Fundamentals of Computer Algorithms”, University Press, 2nd Edition, 2008.
10. Alfred V.Aho, John E Hopcroft, Jeffrey D. Ullman, “Data Structures and Algorithms”, Pearson Education, Reprint 2006.

2402CA103

COMPUTER COMMUNICATION AND NETWORKS

L	T	P	C
3	0	0	3

Prerequisite Computer Organization and Architecture

COURSE OBJECTIVES

1. To explore various data communication techniques.
2. To know network fundamentals and protocols.
3. To understand network addressing and routing concepts.
4. To understand the requirement of reliable and unreliable communication
5. To understand the functionality and concepts of various application layer protocols

COURSE OUTCOMES

At the end of the course, students will be able to

- CO1 Analyze the Communication Model and Data encoding techniques used in Computer Communications
- CO2 Create simple networks by applying networking Protocols
- CO3 Establish Data communication layer in simple networks using Data Link Protocols.
- CO4 Categorize IP addresses using classes of IP and create subnets.
- CO5 Establish Transport layer and application layer in simple networks using protocols.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	2	2	–	–	–	–	2	2
CO2	3	3	3	3	2	–	–	–	3	3
CO3	3	3	3	3	2	–	–	–	3	3
CO4	3	3	2	2	1	–	–	–	2	3
CO5	3	3	3	2	1	–	–	1	3	3
Avg	3	3	2.6	2.4	1.5	-	-	1	2.6	2.8

UNIT I DATA COMMUNICATIONS

8 Hours

Data communications and Networking: Communication model, Data transmission concepts and terminology, Transmission media, Data encoding techniques – Digital data communication techniques: Error detection and correction, Line configurations – Multiplexing: FDM, TDM, Statistical TDM.

UNIT II NETWORK FUNDAMENTALS

8 Hours

Network Architecture: The OSI model, TCP/IP model – Network interface layer: Framing – Reliable transmission: stop and wait protocol, sliding window protocols

UNIT III DATA LINK LAYER

10 Hours

Data link control - Flow Control – Error Detection and Error Correction - MAC – Ethernet, Token ring, Wireless LAN MAC – Blue Tooth – Bridges – Spanning Tree Algorithm.

UNIT IV NETWORK LAYER

9 Hours

Network layer functions – circuit switching – packet switching – IP datagram – IPv4 – Sub netting and classless addressing – IPv6 – ARP – Routing protocols: distance vector, link state – ICMP – ICMPv6 – Case study on Network Design.

UNIT V TRANSPORT LAYER AND APPLICATION LAYER

10 Hours

Transport Layer: Duties of transport layer– User Datagram Protocol – Transmission Control Protocol – Congestion – Congestion control. Application Layer: Application layer Protocols – World Wide Web and HTTP – FTP – Domain name system– Telnet – Electronic mail protocols –SNMP – Case study on Software Defined Networks.

TOTAL : 45 PERIODS

1. Larry L. Peterson and Bruce S. Davie, “Computer Networks: A Systems Approach”, Fifth Edition, Morgan Kaufmann Publishers, 2012.
2. William Stallings, “Data and Computer Communications”, Tenth Edition, Pearson, 2013
3. James F. Kurose, Keith W. Ross, “Computer Networking: A Top-Down Approach”, Fifth Edition, Pearson Education, 2012.
4. Forouzan, “Data Communication and Networking”, Fifth Edition, TMH, 2012.
5. Andrew S. Tannenbaum and David J. Wetherall, “Computer Networks”, Fifth Edition, Pearson Education, 2011.
6. <https://www.javatpoint.com/computer-network-tutorial>

7. https://www.tutorialspoint.com/data_communication_computer_network/index.htm
8. <https://www.geeksforgeeks.org/computer-network-tutorials/>
9. <https://www.tutorialsworld.com/ns2/NS2-1.htm>
10. Forouzan, “Data Communication and Networking”, Fifth Edition, TMH, 2012.

2402CA104

HUMAN RESOURCE MANAGEMENT

L	T	P	C
3	0	0	3

Prerequisite 1. Software Quality Management 2. Software Project Management

COURSE OBJECTIVES

1. To enable the students to understand the various HR functions in-depth.
2. To familiarize students with contemporary practices.
3. To enable the students to understand the challenges in domestic and HRM

COURSE OUTCOMES

- On the successful completion of the course, students will be able to
- CO1 Explain the importance of human resource and their effective management in organization.
- CO2 Describe the procedures and practices using recruitment and selection
- CO3 Summarize how to implement the successful training program using types of training methods.
- CO4 Classify the employee benefit packages with compensation plans.
- CO5 Associate the activities involved in Performance Appraisal by discussing challenges and implications in grievance redressal methods.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	2	1	–	2	1	–	2	–	–
CO2	2	2	1	–	2	2	–	2	–	–
CO3	2	2	1	–	2	1	–	3	–	–
CO4	2	2	–	–	2	1	–	2	–	–
CO5	3	3	1	–	3	2	–	3	–	–
Avg	2.2	2.2	1	–	2.2	1.4	–	2.4	–	–

UNIT I PERSPECTIVES IN HUMAN RESOURCE MANAGEMENT 8 Hours

Evolution of human resource management – The importance of the human factor – Challenges – HR functions - Role of human resource manager – Human resource policies – Computer applications in human resource management – Human resource accounting and audit – environment of HRM.

UNIT II THE CONCEPT OF BEST FIT EMPLOYEE 8 Hours

Importance of Human Resource Planning – Forecasting human resource requirement – matching supply and demand - Internal and External sources. Recruitment - Selection – induction – Socialization benefits.

UNIT III TRAINING AND EXECUTIVE DEVELOPMENT 10 Hours

Types of training methods – purpose- benefits- resistance. Executive Development Programme – Common practices - Benefits – Self development – Knowledge management.

UNIT IV SUSTAINING EMPLOYEE INTEREST 9 Hours

Compensation plan – Reward, remuneration, incentives and benefits – Career management – Development of mentor – Protégé relationships.

UNIT V PERFORMANCE EVALUATION AND IHRM 10 Hours

Method of performance evaluation – Feedback – Promotion – Demotion - Transfer and Separation – Implication of job change - The control process – Importance – Methods – grievances – Causes – Implications – Redressal methods.

REFERENCES TOTAL : 45 PERIODS

1. Anne-Wil Harzing, Ashly Pinnington, Human Resource Management, Sage Publication 4th edition, 2017.
2. Dessler & Varkkey, Human Resource Management, 14th edition, Pearson Education Limited, 2016
3. Aswathappa, Human Resource and Personnel Management- Text & Cases, Tata McGraw Hill, 2013
4. Luis R. Gomez-Mejia, David B. Balkin, Robert L. Cardy, Managing Human Resource, PHI Learning, 2012
5. Ivancevich, Human Resource Management, McGraw Hill 2012
6. Bernadin, Human Resource Management, Tata McGraw Hill, 8th edition 2012
7. Uday Kumar Haldar, Juthika Sarkar. Human Resource management. Oxford. 2012.
8. Decenzo and Robbins, Human Resource Management, Wiley, 8th Edition, 2007.
9. Biswajeet Pattanayak, Human Resource Management, PHI, Third Edition, 2005
10. Dr. V.P. Michael, Human Resource Management & Human Relations, Himalaya Publishing House, 2005
11. <https://www.inc.com/encyclopedia/human-resource-management.html>

2402CA105

PYTHON PROGRAMMING

Prerequisite Programming Basics

L	T	P	C
3	0	0	3

COURSE OBJECTIVES

1. To develop Python programs with conditionals, loops and functions.
2. To use Python data structures – lists, tuples, dictionaries.
3. To do input/output with files in Python
4. To use modules, packages and frameworks in python
5. To define a class with attributes and methods in python

COURSE OUTCOMES

On completion of the course the student would be able to

- CO1 Develop Python program using control structures
- CO2 Develop Python program using Python collection data types.
- CO3 Read and write data from/to files in Python Programs
- CO4 Develop Python program using modules and packages
- CO5 Develop python program using object oriented programming concepts

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	2	1	-	-	2	3	2
CO2	3	2	3	2	1	-	-	2	3	2
CO3	3	2	3	3	1	-	-	2	3	3
CO4	3	2	3	3	1	-	-	2	3	3
CO5	3	2	3	3	1	-	-	2	3	2
Avg	3	2	2.8	2.6	1	-	-	2	3	2.4

UNIT I BASICS OF PYTHON

9 Hours

Introduction to Python Programming – Python Interpreter and Interactive Mode– Variables and Identifiers – Arithmetic Operators – Values and Types – Statements. Operators – Boolean Values – Operator Precedence – Expression – Conditionals: If-Else Constructs – Loop Structures/Iterative Statements – While Loop – For Loop – Break Statement- Continue statement – Function Call and Returning Values – Parameter Passing – Local and Global Scope – Recursive Functions

UNIT II DATA TYPES IN PYTHON

9 Hours

Lists, Tuples, Sets, Strings, Dictionary, Modules: Module Loading and Execution – Packages – Making Your Own Module – The Python Standard Libraries.

UNIT III FILE HANDLING AND EXCEPTION HANDLING

9 Hours

Files: Introduction – File Path – Opening and Closing Files – Reading and Writing Files –FilePosition –Exception: Errors and Exceptions, Exception Handling, Multiple Exceptions

UNIT IV MODULES, PACKAGES AND FRAMEWORKS

9 Hours

Modules: Introduction – Module Loading and Execution – Packages – Making Your Own Module –The Python Libraries for data processing, data mining and visualization- NUMPY, Pandas,Matplotlib, Frameworks- -Django, Flask, Web2Py

UNIT V OBJECT ORIENTED PROGRAMMING IN PYTHON

9 Hours

Creating a Class, Class methods, Class Inheritance, Encapsulation, Polymorphism, class methodvs. static methods, Python object persistence

TOTAL : 45 PERIODS

REFERENCES

1. Reema Thareja, “Python Programming using Problem Solving Approach”, Oxford University Press, First edition, 2017
2. Allen B. Downey, “Think Python: How to Think Like a Computer Scientist”, Second Edition, Shroff, O., Reilly Publishers, 2016 (<http://greenteapress.com/wp/thinkpython/>)
3. Guido van Rossum, Fred L. Drake Jr., “An Introduction to Python – Revised and Updated for Python 3.2”, Network Theory Ltd., First edition, 2011
4. John V Guttag, “Introduction to Computation and Programming Using Python”, Revised and Expanded Edition, MIT Press, 2013
5. Charles Dierbach, “Introduction to Computer Science using Python”, Wiley India Edition, First Edition, 2016

2402CA106 OBJECT ORIENTED SOFTWARE ENGINEERING

L	T	P	C
3	0	0	3

Prerequisite Principles of Software Engineering

COURSE OBJECTIVES

1. To understand the phases in object oriented software development
2. To gain fundamental concepts of requirements engineering and analysis.
3. To know about the different approach for object oriented design and its methods
4. To learn about how to perform object oriented testing and how to maintain software
5. To provide various quality metrics and to ensure risk management.

COURSE OUTCOMES

- At the end of the course, students will be able to
- CO1 Design object oriented software using appropriate process models
 - CO2 Design and Develop UML diagrams for software projects
 - CO3 Apply Design Patterns for a software process.
 - CO4 Categorize testing methods and compare different testing tools for software processes.
 - CO5 Analyze object oriented metrics and quality for software engineering processes

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	3	2	2	1	-	2	3	2
CO2	3	3	3	3	2	1	-	2	3	2
CO3	3	3	3	3	2	1	-	2	3	2
CO4	3	3	3	3	1	1	-	2	3	3
CO5	3	3	2	3	1	1	-	2	3	3
Avg	3	3	2.8	2.8	1.6	1	-	2	3	2.4

UNIT I SOFTWARE DEVELOPMENT AND PROCESS MODELS

9 Hours

Introduction to Software Development – Challenges – An Engineering Perspective – Object Orientation – Software Development Process – Iterative Development Process – Process Models– Life Cycle Models – Unified Process – Iterative and Incremental – Agile Processes

UNIT II MODELING OO SYSTEMS

9 Hours

Object Oriented Analysis (OOA / Coad-Yourdon), Object Oriented Design (OOD/Booch), Hierarchical Object Oriented Design (HOOD), Object Modeling Technique (OMT) – Requirement Elicitation – Use Cases – SRS Document – OOA - Identification of Classes and Relationships, Identifying State and Behavior – OOD - Interaction Diagrams – Sequence Diagram – Collaboration Diagrams - Unified Modeling Language and Tools.

UNIT III DESIGN PATTERNS

9 Hours

Design Principles – Design Patterns – GRASP – GoF – Dynamic Object Modeling – Static Object Modeling

UNIT IV SYSTEM TESTING

9 Hours

Software testing: Software Verification Techniques – Object Oriented Checklist :- Functional Testing – Structural Testing – Class Testing – Mutation Testing – Levels of Testing – Static and Dynamic Testing Tools - Software Maintenance – Categories – Challenges of Software Maintenance – Maintenance of Object Oriented Software – Regression Testing

UNIT V SOFTWARE QUALITY AND METRICS

9 Hours

Need of Object Oriented Software Estimation – Lorenz and Kidd Estimation – Use Case Points Method – Class Point Method – Object Oriented Function Point – Risk Management – Software Quality Models – Analyzing the Metric Data – Metrics for Measuring Size and Structure – Measuring Software Quality - Object Oriented Metrics

TOTAL : 45 PERIODS

REFERENCES

1. Yogesh Singh, Ruchika Malhotra, “ Object – Oriented Software Engineering”, PHI Learning Private Limited ,First edition,2012
2. Ivar Jacobson. Magnus Christerson, Patrik Jonsson, Gunnar Overgaard, “Object Oriented Software Engineering, A Use Case Driven Approach”, Pearson Education, Seventh Impression, 2009
3. Craig Larman, “Applying UML and Patterns, an Introduction to Object-Oriented Analysis and Design and Iterative Development”, Pearson Education, Third Edition, 2008.
4. Grady Booch, Robert A. Maksimchuk, Michael W. Engle, Bobbi J. Young, Jim Conallen,

2402CA107 ADVANCED DATA STRUCTURES AND ALGORITHMS LABORATORY

L	T	P	C
0	0	4	2

Prerequisite C Programming

COURSE OBJECTIVES

1. To acquire the knowledge of using advanced tree structures
2. To learn the usage of heap structures
3. To understand the usage of graph structures and spanning trees
4. To understand the problems such as matrix chain multiplication, activity selection and Huffman coding
5. To understand the necessary mathematical abstraction to solve problems.

COURSE OUTCOMES

- At the end of the course, students will be able to
- CO1 Design and implement basic and advanced data structures extensively
 - CO2 Design algorithms using graph structures
 - CO3 Design and develop efficient algorithms with minimum complexity using design techniques
 - CO4 Develop programs using various algorithms.
 - CO5 Choose appropriate data structures and algorithms, understand the ADT/libraries, and use it to design algorithms for a specific problem

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	3	2	1	-	-	1	3	2
CO2	3	3	3	2	1	-	-	1	3	2
CO3	3	3	3	3	1	-	-	2	3	3
CO4	3	2	2	3	1	-	-	2	3	3
CO5	3	3	3	3	1	-	-	2	3	3
Avg	3	2.6	2.8	2.6	1	-	-	1.6	3	2.6

List of Experiments

1. Stack ADT implementation with arrays and linked lists.
2. Queue ADT implementation with arrays and linked lists
3. Implementation of Merge Sort and Quick Sort
4. Implementation of a Binary Search Tree : Red-Black Tree implementation
5. Heap Implementation
6. Fibonacci Heap Implementation : Graph Traversals
7. Spanning Tree Implementation
8. Shortest Path Algorithms (Dijkstra's algorithm, Bellman Ford Algorithm)
9. Implementation of Matrix Chain Multiplication
10. Activity Selection and Huffman Coding Implementation

TOTAL : 60 PERIODS

REFERENCES

1. Lipschutz Seymour, "Data Structures Schaum's Outlines Series", Tata McGraw Hill, 3rd Edition, 2014.
2. Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman, "Data Structures and Algorithms", Pearson Education, Reprint 2006.
3. <http://www.coursera.org/specializations/data-structures-algorithms>
4. http://www.tutorialspoint.com/data_structures_algorithms
5. <http://www.geeksforgeeks.org/data-structures/>

2402CA108

PYTHON PROGRAMMING LABORATORY

Prerequisite C Programming

L	T	P	C
0	0	4	2

COURSE OBJECTIVES

1. Develop Python programs with conditionals, loops and functions
2. Represent compound data using Python lists, tuples, dictionaries
3. Read and write data from/to files in Python
4. Implement NumPy, Pandas, Matplotlib libraries
5. Implement object oriented concepts

COURSE OUTCOMES

- At the end of the course, students will be able to
- CO1 Apply Python syntax, control structures, and functions to solve problems in mathematics, science, and general applications.
 - CO2 Use core Python data structures such as lists, tuples, dictionaries, and sets to store, process, and manage data.
 - CO3 Implement file handling techniques to perform read and write operations for data management tasks.
 - CO4 Utilize Python libraries (NumPy, Pandas, Matplotlib, SciPy) to perform computational data visualization
 - CO5 Develop Python programs incorporating exception handling and object-oriented programming concepts for real-time applications.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	2	2	2	1	-	-	2	3	2
CO2	3	3	2	2	1	-	-	2	3	2
CO3	2	2	2	3	1	-	-	1	2	2
CO4	2	2	3	3	2	-	-	2	3	2
CO5	2	3	3	3	2	-	-	3	3	3
Avg	2.4	2.4	2.4	2.6	1.4	-	-	2	2.8	2.2

LIST OF EXPERIMENTS:

Note: The examples suggested in each experiment are only indicative. The lab instructor is expected to design other problems on similar lines.

1. Python programming using simple statements and expressions (exchange the values of two variables, circulate the values of n variables, distance between two points).
2. Scientific problems using Conditionals and Iterative loops.
3. Implementing applications using Lists, Tuples.
4. Implementing applications using Sets, Dictionaries.
5. Implementing programs using Functions.
6. Implementing programs using Strings.
7. Implementing programs using written modules and Python Standard Libraries (pandas, numpy, Matplotlib, scipy)
8. Implementing real-time/technical applications using File handling.
9. Implementing real-time/technical applications using Exception handling.
10. Creating and Instantiating classes

TOTAL : 60 PERIODS

REFERENCES

1. Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", Second Edition, Updated for Python 3, Shroff/O'Reilly Publishers, 2016.
2. Shroff "Learning Python: Powerful Object-Oriented Programming; Fifth edition, 2013.
3. David M. Beazley "Python Essential Reference". Addison-Wesley Professional; Fourth edition, 2009.
4. David M. Beazley "Python Cookbook" O'Reilly Media; Third edition (June 1, 2013).
5. <http://www.edx.org/>

2402CA109

Professional Development Course -I

L	T	P	C
0	0	2	1

Prerequisite Basic Communication Skills

COURSE OBJECTIVES

1. Understand and develop personal and professional traits like personality, etiquette, emotional intelligence, and employability skills.
2. Enhance verbal communication through grammar, vocabulary, and structured sentence building.
3. Apply logical reasoning and analytical thinking to solve problems effectively.

COURSE OUTCOMES

At the end of the course, students will be able to

- CO1 Apply personality development and communication skills for self-presentation and group interaction.
- CO2 Use verbal ability and grammar to improve business communication
- CO3 Solve problems using logical reasoning and decision-making techniques.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	1	1	–	–	3	–	–	3	–	–
CO2	1	1	–	–	3	–	–	2	–	–
CO3	2	3	1	–	2	–	–	2	–	–
Avg	1.3	1.6	1	-	2.6	-	-	2.3	–	–

UNIT I Personality and Essentials 10 Hours

Goal setting, Emotional Intelligence, Etiquettes – Dressing etiquettes, Telephone etiquettes, Employability Skills – Hard Skills vs. Soft Skills, Resume building – Creative resume, Self- introduction – Formal/Informal Self introduction, Self-introduction practice, Group Discussion - Mock GD

UNIT II Verbal Ability 10 Hours

SVA, Tenses, Preposition, Business Vocabulary, Art of reading, Art of Conversation, Sentence building, Question making

UNIT III Logical Reasoning 10 Hours

Coding and Decoding- Direction Sense - Blood Relations - Non Verbal Reasoning - Logical Deduction - Decision making

Reference

TOTAL : 30 PERIODS

1. Soft Skills: Know Yourself & Know the World – K. Alex, S. Chand & Company Ltd.
2. Personality Development and Soft Skills – Barun K. Mitra, Oxford University Press.
3. Communication Skills – Sanjay Kumar and Pushp Lata, Oxford University Press.
4. The 7 Habits of Highly Effective People – Stephen R. Covey, Simon & Schuster.
5. Developing Soft Skills for Corporate Readiness – G. S. Gupta, PHI Learning Pvt. Ltd.

BRIDGE COURSES

Bridge Course (For the M.C.A students admitted under non-computer-science background category)						
Course Code	Course Name	L	T	P	C	Category
SEMESTER I						
2401CA001	Problem Solving and Programming in C	3	0	2	4	PC
2401CA002	Data Structures and Algorithms	3	0	0	3	PC
	TOTAL	6	0	2	7	

2401CA001	PROBLEM SOLVING AND PROGRAMMING IN C	L	T	P	C
Prerequisite C Programming		3	0	2	4

COURSE OBJECTIVES

1. To understand the basic concepts of problem solving approaches and to develop the algorithms
2. Apply the techniques of structured (functional) decomposition to break a program into smaller pieces and describe the mechanics of parameter passing
3. To design, implements, test, and apply the basic C programming concepts

COURSE OUTCOMES

At the end of the course, students will be able to

- CO1 Able to design a computational solution for a given problem.
 CO2 Able to break a problem into logical modules that can be solved (programmed).
 CO3 Able to transform a problem solution into programs involving programming constructs.
 CO4 To write programs using structures, strings, arrays, pointers and files for solving complex computational problems.
 CO5 Able to introduce modularity using functions and pointers which permit ad hoc runtime polymorphism.

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	2	2	1	–	–	1	3	2
CO2	3	3	2	2	1	–	–	1	3	2
CO3	3	2	2	3	1	–	–	1	3	2
CO4	3	2	3	3	1	–	–	1	3	3
CO5	3	2	3	3	1	–	–	1	3	2
Avg	3	2.4	2.4	2.6	1	–	–	1	3	2.2

UNIT I **INTRODUCTION TO COMPUTER PROBLEM SOLVING** 15 Hours

Introduction – The Problem Solving aspect – Top down design – Implementation of algorithm – Program Verification
 – The efficiency of algorithms – The analysis of algorithms – Fundamental Algorithms

UNIT II **FUNDAMENTALS OF C LANGUAGE** 15 Hours

Overview of C – Constants – Variables and Data Types – Operators and Expressions – Managing Input/Output Operations – Formatted I/O – Decision Making – Branching –IF – Nested IF – Switch – Goto – Looping Statements

Lab Experiments:

1. Write programs to get some input , perform some operation and display the output using I/O statements
2. Write a program to execute some specific statements based on the test condition
3. Write programs to implement nested loop

UNIT III **ARRAYS - FUNCTIONS - STRUCTURES AND UNIONS** 15 Hours

Arrays – Character arrays and Strings – String handling Functions – User defined Functions – Categories of Functions – Recursion – Structures and Unions – Array of Structures – Structures and Functions

Lab Experiments

1. Write a program in C to get the largest element of an array using the function.
2. Display all prime numbers between two intervals using functions.
3. Reverse a sentence using recursion.
4. Write a C program to concatenate two strings

UNIT IV **POINTERS** 15 Hours

pointers – pointer operators - storage classes - uses of pointers- arrays and pointers – pointers and strings - pointer indirection pointers to functions - Dynamic memory allocation.

Lab Experiments:

1. Write a C program to Store Student Information in Structure and Display it.
2. The annual examination is conducted for 10 students for five subjects.

UNIT V FILES

15 Hours

File Management in C – Files and Streams – File handling functions – Sequential access file- Random access file – Command line arguments.

Lab Experiments:

1. Write a program to read the data from a file and determine the following: Total marks obtained by each student; Topper of the class

TOTAL : 75 PERIODS

REFERENCES

1. Deitel and Deitel, “C How to Program”, Pearson Education. 2013, 7th Edition
2. Byron S Gottfried, —Programming with C, Schaums Outlines, Second Edition, Tata McGraw-Hill, 2006
3. Brian W. Kernighan and Dennis M. Ritchie, “The C programming Language”, Edition? 2nd edition 2015, Pearson Education India
4. How to solve it by Computer, R. G. Dromey, Pearson education, Fifth Edition, 2007
5. Kamthane, A.N., “Programming with ANSI and Turbo C”, Pearson Education, Delhi, 3rd Edition, 2015 Pradip Dey, Manas Ghosh, —Computer Fundamentals and Programming in C, Second Edition, Oxford University Press, 2013.

401CA002 DATA STRUCTURES AND ALGORITHMS

L	T	P	C
3	0	2	4

Prerequisite C Programming

COURSE OBJECTIVES

1. Be familiar with basic techniques of algorithm analysis.
2. Be exposed to the concept of ADTs.
3. Learn linear data structures-List, Stack and Queue.
4. Learn nonlinear data structures-Tree and Graphs.
5. Be exposed to sorting, searching and hashing algorithms

COURSE OUTCOMES

Upon Completion of the course, the students will be able to

- CO1 Analyze algorithms and determines their time complexity.
 CO2 understand the concepts of data types, data structures and linear structures
 CO3 apply data structures to solve various problems
 CO4 apply different Sorting, Searching and Hashing algorithms
 CO5 understand non-linear data structures

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	3	3	2	2	1	-	-	1	3	2
CO2	3	2	2	2	1	-	-	1	3	2
CO3	3	3	3	3	1	-	-	1	3	2
CO4	3	2	3	3	1	-	-	1	3	2
CO5	3	2	2	2	1	-	-	1	3	2
Avg	3	2.4	2.4	2.4	1	-	-	1	3	2

UNIT I INTRODUCTION

9 Hours

Introduction – The Problem Solving aspect – Top down design – Implementation of algorithm – Program Verification
 – The efficiency of algorithms – The analysis of algorithms – Fundamental Algorithms

UNIT II LINEAR DATA STRUCTURES - STACK, QUEUE

9 Hours

Stack ADT – Operations on Stack - Applications of stack - Queue ADT – Operations on Queue - Applications of Queue.

UNIT III ARRAYS - FUNCTIONS - STRUCTURES AND UNIONS

9 Hours

List ADT - Array-based Implementation - Linked list implementation - Singly Linked Lists - Applications of linked list

UNIT IV POINTERS

9 Hours

Searching: Linear search – Binary Search- comparison of linear search and binary search, Sorting algorithms: Insertion sort - Bubble sort – selection sort

UNIT V FILES

9 Hours

Trees and its representation - Binary Tree – Binary tree traversals — Binary Search Tree - Graphs and its representation - Graph Traversals - Depth-first traversal – breadth-first traversal-Application of graphs

TOTAL : 45 PERIODS

REFERENCES

1. Anany Levitin “Introduction to the Design and Analysis of Algorithms” 3rd Edition, Pearson Education
2. A.K. Sharma, “Data Structures using C”, 2nd Edition, Pearson Education Asia, 2013
3. E.Horowitz, Anderson-Freed and S.Sahni, “Fundamentals of Data structures in C”, 2nd Edition, University Press, 2007
4. E.Balagursamy, “Data Structures using C”, Tata McGraw Hill 2015 Reprint
5. Mark Allen Weiss, “Data Structures and Algorithm Analysis in C”, 2nd Edition, Pearson Education, India, 2016
6. Jean Paul Tremblay and Paul G. Sorensen, “An Introduction to Data Structures with Applications”, 2nd Edition, Tata McGraw Hill, New Delhi, 2017.

LIST OF AUDIT COURSES (AC) Registration for any of these courses is optional to students						
Course Code	Course Name	L	T	P	C	Category
2401AU001	English for Research Paper Writing	2	0	0	0	AC
2401AU002	Disaster Management	2	0	0	0	AC
2401AU003	Value Education	2	0	0	0	AC
2401AU004	Constitution of India	2	0	0	0	AC

2401AU001

English for Research Paper Writing

L T P C

Prerequisite Basic proficiency in English grammar and academic writing skills.

2 0 0 0

COURSE OBJECTIVES

1. Teach how to improve writing skills and level of readability
2. Tell about what to write in each section
3. Summarize the skills needed when writing a Title
4. Infer the skills needed when writing the Conclusion
5. Ensure the quality of paper at very first-time submission

COURSE OUTCOMES

At the end of the course, students will be able to

- CO1 Understand that how to improve your writing skills and level of readability
 CO2 Learn about what to write in each section
 CO3 Understand the skills needed when writing a Title
 CO4 Understand the skills needed when writing the Conclusion
 CO5 Ensure the good quality of paper at very first-time submission

COs Vs POs MAPPING

COs \ POs / PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PSO1	PSO2
CO1	2	2	1	1	2	-	1	3	1	1
CO2	2	2	1	1	2	-	1	3	1	1
CO3	2	1	1	1	2	-	1	3	1	1
CO4	2	1	1	1	2	-	1	3	1	1
CO5	2	2	2	1	2	1	1	3	2	1
Avg	2	1.6	1.2	1	2	1	1	3	1.2	1

UNIT I INTRODUCTION TO RESEARCH PAPER WRITING

6 Hours

Planning and Preparation, Word Order, Breaking up long sentences, Structuring Paragraphs and Sentences, Being Concise and Removing Redundancy, Avoiding Ambiguity and Vagueness

UNIT II PRESENTATION SKILLS

6 Hours

Clarifying Who Did What, Highlighting Your Findings, Hedging and Criticizing, Paraphrasing and Plagiarism, Sections of a Paper, Abstracts, Introduction

UNIT III TITLE WRITING SKILLS

6 Hours

Key skills are needed when writing a Title, key skills are needed when writing an Abstract, key skills are needed when writing an Introduction, skills needed when writing a Review of the Literature, Methods, Results, Discussion, Conclusions, The Final Check

UNIT IV SYSTEM TESTING

6 Hours

Skills are needed when writing the Methods, skills needed when writing the Results, skills are needed when writing the Discussion, skills are needed when writing the Conclusions

UNIT V SOFTWARE QUALITY AND METRICS

6 Hours

Useful phrases, checking Plagiarism, how to ensure paper is as good as it could possibly be the first- time submission

TOTAL : 30 PERIODS

REFERENCES

1. Adrian Wallwork , English for Writing Research Papers, Springer New York DordrechtHeidelberg London, 2011
2. Day R How to Write and Publish a Scientific Paper, Cambridge University Press 2006
3. Goldbort R Writing for Science, Yale University Press (available on Google Books) 2006
4. Highman N, Handbook of Writing for the Mathematical Sciences, SIAM. Highman's book1998.